WORKED SOLUTION

# STEP III 1991, Question 15

Probability and Statistics • Counting Pairs in Samples

**STEP III 1991, Question 15**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q15

## The key idea

Every set of $k$ cards is equally likely, so each probability is a count of hands divided by $\binom{2n}{k}$. To count hands with a given pair structure, build them in stages. First choose which sorts appear as complete pairs. Then choose which other sorts appear once. Finally choose which of the two cards of each single sort we hold. Once that count is in place, every part of the question is a special case of one formula.

## Solution

Think of the $2n$ cards as all distinguishable (the two cards of a sort might be numbered 1 and 2). Drawing $k$ cards without replacement then gives each of the $\binom{2n}{k}$ subsets of size $k$ the same probability. The hand contains a *pair* of a sort when both cards of that sort are drawn.

### Four cards, no pairs

A hand with no pair contains four different sorts, one card of each. There are $\binom{n}{4}$ ways to choose the sorts and 2 ways to choose the card of each sort, which gives $2^4\binom{n}{4}$ hands. Since

$$\binom{2n}{4} = \frac{2n(2n-1)(2n-2)(2n-3)}{24} = \frac{4n(n-1)(2n-1)(2n-3)}{24},$$

we get

$$\mathrm{P}(\text{no pairs}) = \frac{16 \cdot n(n-1)(n-2)(n-3)/24}{4n(n-1)(2n-1)(2n-3)/24} = \frac{4(n-2)(n-3)}{(2n-1)(2n-3)},$$

as required. (Here $n \geq 4$ makes sure four different sorts exist.)

**Four cards, exactly one pair**

Choose the sort that appears as a pair ($n$ ways). The other two cards must come from two different sorts out of the remaining $n-1$, which can be chosen in $\binom{n-1}{2}$ ways, with 2 choices of card for each. The number of hands is

$$n \cdot \frac{(n-1)(n-2)}{2} \cdot 4 = 2n(n-1)(n-2),$$

so

$$\mathrm{P}(\text{exactly one pair}) = \frac{24 \cdot 2n(n-1)(n-2)}{4n(n-1)(2n-1)(2n-3)} = \frac{12(n-2)}{(2n-1)(2n-3)}.$$

**$k$ cards, exactly $r$ pairs**

Suppose the hand has exactly $r$ pairs. They use $2r$ cards. The remaining $k-2r$ cards all belong to different sorts, none of which is one of the $r$ paired sorts, since otherwise we would have more pairs. So we count:

- the paired sorts: $\binom{n}{r}$ ways;
- the $k-2r$ sorts appearing once, chosen from the other $n-r$ sorts: $\binom{n-r}{k-2r}$ ways;
- which of the two cards we hold of each single sort: $2^{k-2r}$ ways.

Each hand with exactly $r$ pairs arises from exactly one such sequence of choices. Therefore

$$\boxed{\mathrm{P}(\text{exactly } r \text{ pairs}) = \frac{2^{k-2r}\binom{n}{r}\binom{n-r}{k-2r}}{\binom{2n}{k}}}$$

? **Ask yourself** Is this ever zero, and does it agree with the four-card answers?

The single sorts need $k-2r \le n-r$, that is, $r \ge k-n$. If $k > n$, any $k$ cards must contain at least $k-n$ pairs, and $\binom{n-r}{k-2r} = 0$ handles this automatically. With $k=4$, the case $r=0$ gives $2^4\binom{n}{4}/\binom{2n}{4}$ and $r=1$ gives $2^2\, n\binom{n-1}{2}/\binom{2n}{4}$, which are exactly the two counts above.

**$k$ even, all cards in pairs**

Put $k = 2m$. The hand consists of $m$ pairs exactly when it is the union of $m$ complete sorts. There are $\binom{n}{m}$ such hands (this is the general formula with $r = m$). So

$$\mathrm{P} = \frac{\binom{n}{m}}{\binom{2n}{2m}} = \frac{n!}{m!\,(n-m)!} \cdot \frac{(2m)!\,(2n-2m)!}{(2n)!}.$$

To simplify, pair up the even and odd factors of a factorial:

$$(2j)! = \big(2 \cdot 4 \cdots 2j\big)\big(1 \cdot 3 \cdots (2j-1)\big) = 2^j\, j!\,\big(1 \cdot 3 \cdots (2j-1)\big).$$

Write $O_j = 1 \cdot 3 \cdots (2j-1)$, with $O_0 = 1$. Then $(2j)!/j! = 2^j O_j$, so

$$\mathrm{P} = \frac{2^m O_m \cdot 2^{n-m} O_{n-m}}{2^n O_n} = \frac{O_m\, O_{n-m}}{O_n}.$$

Dividing $O_n$ by $O_{n-m}$ leaves the odd numbers from $2n-2m+1 = 2n-k+1$ up to $2n-1$, so

$$\mathrm{P}(\tfrac{1}{2}k \text{ pairs}) = \frac{1 \times 3 \times 5 \times \cdots \times (k-1)}{(2n-1)(2n-3)\cdots(2n-k+1)},$$

as required.

There's also a neat direct argument. Shuffle the whole pack and let the hand be the top $k$ cards. Look at the top card. Its partner is equally likely to be in any of the other $2n-1$ positions, and $k-1$ of those are in the hand, so the partner is in the hand with probability $\frac{k-1}{2n-1}$. If it is, remove that pair. What remains is a random arrangement of $2n-2$ cards with the hand being the top $k-2$ of them. Repeating the argument gives $\frac{k-3}{2n-3}$, and so on down to $\frac{1}{2n-k+1}$. Multiplying these conditional probabilities gives the same product.

## Checks

- For four cards the three cases are 0, 1 and 2 pairs, and 2 pairs has probability $\binom{n}{2}/\binom{2n}{4} = \frac{3}{(2n-1)(2n-3)}$. Indeed

  $$4(n-2)(n-3) + 12(n-2) + 3 = 4n^2 - 8n + 3 = (2n-1)(2n-3),$$

  so the three probabilities add to 1.
- With $n = 5$ and four cards the probabilities are $\frac{8}{21}$, $\frac{4}{7}$ and $\frac{1}{21}$. A computer enumeration of all hands for $4 \le n \le 8$ and every $k$ from 3 to $2n-1$ agreed with the general formula and with the all-pairs formula.
- With $k = 2$ the all-pairs formula gives $\frac{1}{2n-1}$, the chance that the second card matches the first.

## Common slips

- Choosing the single cards from all $2n-2r$ remaining cards with $\binom{2n-2r}{k-2r}$. That allows two of them to be of the same sort, which creates extra pairs.
- Forgetting the factor $2^{k-2r}$. Each sort that appears once could be either of its two cards.

## Beyond STEP

The number of hands with exactly $r$ pairs is a typical "choose the structure, then fill it in" count, and the product $1 \cdot 3 \cdots (2m-1)$ (the double factorial $(2m-1)!!$) counts the

ways to split $2m$ objects into pairs. Double factorials turn up again as the even moments of the standard normal distribution. The same matching problem, of how many pairs survive in a random sample, appears in the classical "lost socks" and birthday-type problems. Grimmett and Welsh, *Probability: An Introduction*, gives a good grounding in this kind of counting.

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